Feature Engineering for Longitudinal Alzheimer's Disease Progression Prediction

Technical Description of the Feature Engineering Pipeline

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1 Overview

The feature engineering pipeline transforms per-subject longitudinal visit records into a fixed-width, visit-count-agnostic feature vector suitable for gradient-boosted tree classification.

Each subject in the dataset has between 2 and 10 clinic visits. The raw data encodes repeated measurements as variable-length lists (one entry per visit), which cannot be fed directly to a standard tabular classifier. The pipeline extracts summary statistics from these lists, yielding a single row per subject regardless of the number of visits observed.

The transformation is implemented in `create_delta_features()` and `preprocess_data()`. All feature engineering is applied *independently* to the training and test splits inside the cross-validation loop, so no information leaks from held-out samples.

The final feature set contains three broad categories:

1. **Static features** — demographics and comorbidities measured once or assumed stable across visits (18 variables).
2. **Longitudinal summary features** — visit-agnostic statistics derived from each repeated-measurement variable (up to 13 statistics per longitudinal column).
3. **Engineered interaction features** — explicit cross-variable interaction terms capturing dual sensory impairment (3 variables).

2 Input Variables

2.1 Static Variables

Static variables are collected at baseline or treated as time-invariant. They are passed directly to the model without transformation.

Variable	**Type**	**Description**
Numeric		
`age`	Continuous	Age at baseline visit (years)
`EDUC`	Continuous	Years of formal education
`SMOKYRS`	Continuous	Total lifetime years of smoking
Binary (0/1)		
`NACCFAM`	Binary	Family history of cognitive impairment or dementia
`CVHATT`	Binary	History of myocardial infarction / heart attack
`CVAFIB`	Binary	History of atrial fibrillation
`DIABETES`	Binary	Diabetes mellitus diagnosis
`HYPERCHO`	Binary	Hypercholesterolaemia diagnosis
`HYPERTEN`	Binary	Hypertension diagnosis
`B12DEF`	Binary	Vitamin B_{12} deficiency diagnosis
`DEPD`	Binary	Active depression diagnosis
`ANX`	Binary	Active anxiety diagnosis
`NACCTBI`	Binary	History of traumatic brain injury
`HISPANIC`	Binary	Hispanic/Latino ethnicity

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Variable	Type	Description
Categorical (unordered)		
`SEX`	Categorical	Sex (male / female)
`RACE`	Categorical	Self-reported race
`ALCOHOL`	Categorical	Alcohol consumption level (none / social / moderate / heavy)
`NACCNE4S`	Categorical	APOE $\varepsilon 4$ allele count (0 / 1 / 2 copies)

Categorical variables are encoded as `pandas category` dtype and consumed natively by XGBoost with `enable_categorical=True`, which applies an optimised tree-splitting strategy for unordered categories.

2.2 Longitudinal Variables

Each of the 19 variables below is stored as a list of per-visit measurements. Missing visits within a sequence are represented as `NaN`.

Variable	NACC Domain	Description
Global cognitive and psychiatric scales		
`NACCMMSE`	Cognition	Mini-Mental State Examination score (0–30; lower = worse)
`CDRSUM`	Global staging	CDR® Sum of Boxes (0–18; higher = worse)
`NACCGDS`	Neuropsychiatry	Geriatric Depression Scale short-form score (0–15)
Functional Activities Questionnaire (FAQ) items		
`BILLS`	ADL	Ability to pay bills and handle bank statements
`TAXES`	ADL	Ability to prepare tax returns or handle finances
`SHOPPING`	ADL	Ability to shop alone (groceries, clothing)
`GAMES`	ADL	Ability to play games of skill (cards, chess)
`STOVE`	ADL	Ability to prepare a hot meal safely
`MEALPREP`	ADL	Ability to assemble and serve a complete meal
`EVENTS`	ADL	Ability to keep track of current events
`PAYATTN`	ADL	Ability to pay attention to / understand TV, book, or magazine
`REMDATES`	ADL	Ability to remember appointments, family events, holidays
`TRAVEL`	ADL	Ability to travel out of the neighbourhood
Anthropometric and behavioural		
`NACCBMI`	Physical health	Body mass index at visit
`TOBAC30`	Substance use	Tobacco use in the past 30 days (pack-years proxy)
Social and sensory		
`NACCLIVS`	Social	Living situation (independent / assisted / dependent)
`COMMUN`	Social	Ability to communicate with others
`hearing`	Sensory	Composite hearing impairment score (derived from six NACC var
`vision`	Sensory	Composite vision impairment score (derived from six NACC varia
Temporal reference		
`months_since_baseline`	Follow-up	Elapsed months from the initial visit to each follow-up visit

The `hearing` composite is derived from `HEARING`, `HEARAID`, `HEARWAID` and the `vision` composite from VISION, VISCORR, VISWCORR in preprocessing. Both range from 0 (no impairment) to 1 (severe impairment).

3 Longitudinal Feature Engineering

Let $\mathcal{V}(x) = \{x_i : x_i \neq \texttt{NaN}\}$ denote the set of valid (non-missing) observations for variable x, ordered chronologically, and let $n = |\mathcal{V}(x)|$ be the number of valid visits. Where time is referenced, t_i denotes the elapsed months from baseline corresponding to the i-th valid observation.

3.1 Standard Summary Statistics

The following 9 statistics are computed for every longitudinal variable:

$$\mathit{mean} = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{1}$$

$$\mathit{max} = \max_i x_i \tag{2}$$

$$\mathit{min} = \min_i x_i \tag{3}$$

$$\mathit{std} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2} \quad (n \geq 2,\ \text{else } \texttt{NaN}) \tag{4}$$

$$\mathit{range} = \max_i x_i - \min_i x_i \tag{5}$$

$$\mathit{first} = x_1 \quad \text{(first valid observation)} \tag{6}$$

$$\mathit{last} = x_n \quad \text{(last valid observation)} \tag{7}$$

$$\mathit{last\_minus\_first} = x_n - x_1 \tag{8}$$

$$\mathit{n\_visits} = n \tag{9}$$

3.2 Time-Normalised Slope

A key design decision is that slopes are computed against *real elapsed time* (months), not visit index. This means the slope is expressed in clinical units per month, making it comparable across patients with irregular visit spacing.

$$\mathit{slope} = \hat{\beta}_1 \quad \text{from OLS: } x_i = \hat{\beta}_0 + \hat{\beta}_1 t_i + \varepsilon_i, \quad n \geq 2 \tag{10}$$

When fewer than 2 valid observations exist, *slope* is set to NaN.

3.3 Time-Normalised Acceleration

Acceleration captures the rate of change in the rate of change — i.e., whether a patient is *accelerating* into decline.

Define the pairwise per-month slopes between consecutive valid visits:

$$s_i = \frac{x_{i+1} - x_i}{t_{i+1} - t_i}, \quad i = 1, \ldots, n-1 \tag{11}$$

Acceleration is the OLS slope of s_i regressed on the midpoint times $\tau_i = (t_i + t_{i+1})/2$:

$$acceleration = \hat{\gamma}_1 \quad \text{from OLS: } s_i = \hat{\gamma}_0 + \hat{\gamma}_1 \tau_i + \varepsilon_i \tag{12}$$

Stability requirement. Acceleration is undefined and set to `NaN` for any patient with fewer than 4 valid observations (requiring ≥ 3 pairwise slopes for a meaningful regression). This threshold was raised from 3 to 4 valid visits to mitigate the instability of 2nd-derivative estimates from very short sequences.

3.4 Slope Variability (*std_slope*)

A single linear slope cannot capture non-linear trajectories, step-changes, or periods of temporary recovery. The standard deviation of pairwise per-month slopes quantifies this temporal variability:

$$std\_slope = \text{std}(s_1, s_2, \ldots, s_{n-1}), \quad n \geq 2 \tag{13}$$

where s_i are defined as in Equation (11) (using real elapsed months) and std($\cdot$) is the sample standard deviation (ddof = 1). When $n = 2$ exactly one slope exists, so $std\_slope = 0$. When $n < 2$, $std\_slope =$ `NaN`.

3.5 Maximum Percentage Change (*pct_change*)

Alzheimer's progression can involve discrete step-changes following clinical events. The maximum relative change between consecutive visits captures this:

$$pct\_change = \max_{i\,:\,x_i \neq 0} \left| \frac{x_{i+1} - x_i}{x_i} \right|, \quad n \geq 2 \tag{14}$$

Pairs where $x_i = 0$ are excluded from the maximum to prevent division-by-zero. When all consecutive pairs involve a zero denominator, or $n < 2$, the feature is `NaN`.

3.6 Lagged Observations

Summary statistics such as mean and slope discard temporal ordering; two patients with the same mean and slope but different trajectories ("improved then worsened" vs. "worsened then improved") are indistinguishable. Lagged features resolve this by encoding the most recent visit values explicitly:

$$lag_k(x) = x_{n-k}, \quad k \in \{1, 2, 3\}, \quad n \geq k + 1 \tag{15}$$

So *lag1* is the second-to-last valid observation, *lag2* is the third-to-last, and *lag3* is the fourth-to-last. If insufficient valid observations exist, the lag is `NaN`.

Lags are computed for 7 clinically prioritised variables: `NACCMMSE`, `CDRSUM`, `NACCGDS`, `hearing`, `vision`, `BILLS`, and `SHOPPING`, yielding 21 additional features. These cover the primary cognitive screening instrument, global disease severity, depression, both sensory modalities, and two representative functional activity items.

4 Visit Timing Features

The `months_since_baseline` column encodes the elapsed months from the initial visit to each subsequent visit. Because this variable is the time axis itself, computing a slope or acceleration over it is not meaningful. Instead, two features describing the *spacing* of visits are extracted:

$$\mathit{interval\_mean} = \frac{1}{n-1}\sum_{i=1}^{n-1}(t_{i+1} - t_i), \quad n \geq 2 \tag{16}$$

$$\mathit{interval\_std} = \mathrm{std}(t_2 - t_1,\ t_3 - t_2,\ \ldots,\ t_n - t_{n-1}), \quad n \geq 3 \tag{17}$$

interval_mean is simply the average time between assessments (months). *interval_std* captures the *irregularity* of the visit schedule.

In addition, the standard summary statistics in Section 3.1 are also computed for `months_since_baseline` (mean, max, min, std, range, first, last, last_minus_first, n_visits), so that features such as *months_since_baseline_last* (total follow-up duration) are available to the model.

5 Hearing × Vision Interaction Features

Dual sensory impairment — concurrent hearing and vision loss — is a well-documented risk factor for dementia, and its effect is reported to be multiplicative rather than additive [**?**]. A model that treats hearing and vision as independent inputs cannot capture this synergy. Three explicit interaction terms are therefore computed from the engineered *last* values of each sensory composite:

$$\mathit{hearing\_vision\_product} = \mathrm{hearing\_last} \times \mathrm{vision\_last} \tag{18}$$

$$\mathit{hearing\_vision\_sum} = \mathrm{hearing\_last} + \mathrm{vision\_last} \tag{19}$$

$$\mathit{hearing\_vision\_mean} = \frac{\mathrm{hearing\_last} + \mathrm{vision\_last}}{2} \tag{20}$$

The product term is zero if either sense is unimpaired and positive only when both are impaired, directly encoding the co-occurrence pattern. The sum and mean terms capture overall sensory burden.

6 Full Feature Summary

6.1 Feature Count by Category

The exact count varies slightly depending on which columns are present in a given dataset split, since `preprocess_data` filters to columns that actually exist.

6.2 Per-Longitudinal-Variable Feature Suffixes

For each of the 19 longitudinal variables v, the following named features are created:

Table 3: Number of features by category in the final feature matrix passed to the XGBoost classifier.

Category	**Features per variable**	**Total features**
Static (numeric + binary + categorical)	1	18
Longitudinal summary (19 vars × 11 stats)	11	209
of which: standard summary	9	171
of which: time-normalised slope	1	19
of which: acceleration (≥4 visits)	1	19
Slope variability *std_slope* (19 vars)	1	19
Maximum percentage change *pct_change* (19 vars)	1	19
Visit timing features (`months_since_baseline`)	—	11
standard summary (9 stats)	—	9
interval mean / std	—	2
Lagged features (7 vars × 3 lags)	3	21
Hearing × Vision interactions	—	3
Total (approx.)		**≈ 300**

Suffix	**Formula**	**Min. visits required**
`_mean`	$\bar{x}$	1
`_max`	$\max x_i$	1
`_min`	$\min x_i$	1
`_std`	sample std of x_i	2
`_range`	$\max x_i - \min x_i$	1
`_first`	x_1	1
`_last`	x_n	1
`_last_minus_first`	$x_n - x_1$	1
`_n_visits`	n	1
`_slope`	OLS slope of x on t (months)	2
`_acceleration`	OLS slope of s_i on τ_i	**4**
`_std_slope`	std of s_i; 0 if $n = 2$	2
`_pct_change`	$\max \lvert x_{i+1} - x_i\rvert / \lvert x_i\rvert$	2
`_lag1`	x_{n-1} (selected vars only)	2
`_lag2`	x_{n-2} (selected vars only)	3
`_lag3`	x_{n-3} (selected vars only)	4

6.3 Design Rationale

Visit-count independence. Because patients have differing numbers of visits, all features are summary statistics that collapse the visit dimension. The model therefore generalises across patients with 2 visits and patients with 10 visits without requiring imputation or fixed-length padding.

Time normalisation. All rate features (slope, acceleration, std_slope) use actual elapsed time in months rather than visit index. This ensures that a 10-point MMSE decline over 6 months is correctly distinguished from the same decline over 4 years, as the former is associated with a much more aggressive trajectory.

Missing value handling. Features that require more observations than a patient has (e.g. acceleration requires ≥ 4) are set to `NaN` rather than imputed. XGBoost handles missing values natively through its learnable default branch direction at each split node.

No scaling. The pipeline does not apply any standardisation or normalisation. XGBoost is scale-invariant, and omitting scaling avoids any possibility of test-set information influencing training-set normalisation.

Categorical encoding. The four categorical variables (`SEX`, `RACE`, `ALCOHOL`, `NACCNE4S`) are passed to XGBoost as `pandas category` dtype with `enable_categorical=True`. This activates XGBoost's native categorical split strategy (one-hot-optimal), which is both faster and more statistically appropriate than label encoding for unordered factors.

No data leakage. `create_delta_features()` and `preprocess_data()` are called independently within each cross-validation fold, including each LOO fold. No statistics (means, scalers, imputers) are fit on the combined train+test data.

7 Target Variable Definition

A binary label $y \in \{0, 1\}$ is derived from each subject's `Progression` sequence — a list of visit-level diagnostic codes where 0 = Cognitively Normal (CN), 1 = Mild Cognitive Impairment (MCI), and 2 = Alzheimer's Disease (AD).

$$y = \begin{cases} 1 & \text{if } 2 \in \text{Progression} \quad \text{(progression type `AD')} \\ 1 & \text{if } \exists\, v \in \text{Progression} : v > 0 \quad \text{(progression type `CN')} \\ 0 & \text{otherwise} \end{cases} \tag{21}$$

For the **CN cohort** (CN $\rightarrow$ MCI prediction), $y = 1$ if the subject ever progressed beyond cognitively normal. For the **MCI/AD cohort** (MCI $\rightarrow$ AD prediction), $y = 1$ if the subject was ever diagnosed with AD.

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